

Unique continuation for elliptic equations with nonsmooth coefficients

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Abstract

Let P be a differential operator and let V and W be measurable functions. We consider sufficiently regular functions u which satisfy

$$|Pu| \leq V|u| + W|\nabla u|$$

in a ball. We say the triple (P, V, W) has the strong unique continuation property (SUCP) if u vanishes if it vanishes of infinite order at one point.

Theorem (Koch, Tataru): Let $n > 2$ and let P be an elliptic partial differential operator of second order with Lipschitz continuous coefficients. The triple (P, V, W) has the SUCP if $V \in L^{n/2}$ and $W \in L^{n+\epsilon}$.

The talk explains this result. It will give a sketch of the proof and explains its relation to questions occurring in the context of nonlinear dispersive equations.

The talk is based on joint work with D. Tataru.

Original document

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